

Kaan Osmanagaoglu

SOFTWARE ENGINEER | JAVA/KOTLIN + C/C++/PYTHON | DEFENCE + SECURE SYSTEMS

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PROFILE

Software engineer with 9+ years across defence-adjacent robotics, cryptography, secure payments, banking and public-utility systems. Strongest where Java/Kotlin application architecture meets C/C++/Python, Linux-oriented tooling, networking, test automation and full-lifecycle delivery. Earlier work includes RSA BSAFE cryptography and Praesidium Global UGV control/networking for Australian defence work.

SYSTEMS PROOF

- RSA: C/C++/ASM encryption-library work on RSA BSAFE, with Python buildbots/pipelines, HSM integration and validation.
- Praesidium Global: UGV control architecture across C/Python networking, ROS, Lidar/sensor communications and operator HMI.
- Quest + Zeller: certified payment-terminal software: PayWave, Tap-to-Pay, PCI, secure key handling, encryption and offline flows.
- Java/Kotlin delivery: banking, public transport, fuel payments and terminal products where reliability and release discipline mattered.

ROLE MATCH

LANGUAGES

Java/Kotlin, C/C++, Python, Bash and ASM; procedural and object-oriented codebases.

ENGINEERING PROCESS

Architecture decisions, Git configuration management, CI/buildbots, test automation, release support, maintenance and systems integration.

SELECTED SYSTEMS EXPERIENCE

RSA (EMC/Dell)

Software Engineer - Cryptography Libraries

Brisbane | 2016

- Developed RSA BSAFE encryption-library components in C/C++/ASM, with Python automation around buildbots, pipelines and repeatable validation.
- Worked across HSM integration, networking/security surfaces and Android mobile-security PoCs where low-level correctness and traceability mattered.

Praesidium Global / Defence

Robotics and HMI Software Engineer

Sunshine Coast | 2018-2019

- Architected UGV control software across operator tablet HMI, backend communications, sensor networking, ROS and Lidar/collision-detection surfaces.
- Built C/Python networking and integration layers between vehicle systems, control UI and backend services alongside electronics/mechanical engineers.
- Owned architecture, implementation, integration and field-readiness instead of isolated feature delivery.

Quest Payment Systems

Senior Software Engineer

Melbourne | 2023

- Delivered Android and management software across PoS/payment-terminal products with Westpac, NAB and Officeworks workflows.
- Handled PayWave/payment certification constraints, secure key handling, encryption, offline flows and release readiness.

Zeller

Senior Mobile Engineer

Melbourne | 2023-2024

- Served as architect on EFTPOS/payment-terminal products; delivered Tap-to-Pay with shared Kotlin mobile code and React Native bridge integration.
- Modernised terminal architecture from RxJava/MVVM toward Coroutines, MVI and Compose while expanding parallel Maestro E2E coverage.

❖ EDUCATION

Bachelor of Science (Information Technology, Networks and Software Engineering), Queensland University of Technology, 2015.

❖ RESEARCH

- Research Assistant (2014): one-time signature trees for post-quantum cryptography.
- Research Assistant (2015): GPS/GNSS positioning accuracy using real-time kinematics.

❖ ADDITIONAL DELIVERY

Tip.mi

Senior Mobile Developer (Contract)

Melbourne | 2025-2026

- Built greenfield Kotlin/Compose Android and Desktop MVP surfaces with CI/CD, secure signing, staged rollouts and Grafana/Mixpanel telemetry.

Crypto.com

Lead Android Developer

Melbourne | 2022-2023

- Led DeFi Wallet work across staking and cross-chain bridge flows; reviewed protocol papers and coordinated backend/blockchain integrations.

Coles

Senior Android Developer (Contract)

Melbourne | 2021

- Built phone-based fuel-pump payment flows from the ground up, integrating maps, geofencing, PayWave and multiple gateways.

❖ EARLIER JAVA / HMI WORK

PTV / ARQ Group + Suncorp Bank

Public transport and private-banking systems

2017-2021

Delivered Java/Kotlin public transport and banking systems: train-position animations, identity/API integration, Dagger/Retrofit/MVVM and testing practices.

Grabba + Mobey + Modanisa/Huawei

Device, logistics, e-commerce and platform integrations

2016-2021

Built biometric, logistics, e-commerce and mobile-platform software spanning device integrations, map/vector animation, payment gateways and international platform APIs.

❖ ENGINEERING RANGE

LANGUAGES

Java/Kotlin, C/C++, Python, Bash, ASM, procedural and OOP codebases

SYSTEMS

Linux-oriented tooling, networking, ROS, Lidar/sensors, HMI/control interfaces

SECURITY

Cryptography libraries, HSMs, secure key handling, PCI, PayWave, wallets

DELIVERY

Git, CI/CD, buildbots, test automation, release support, maintenance

ARCHITECTURE

Design patterns, modularisation, systems integration, state-driven interfaces

INTERFACES

REST, gRPC, GraphQL, WebSockets, payment/device/backend integrations